

[34255][34256] LI Artificial Intelligence 2

Introduction to Computer Vision

Dr Hyung Jin Chang

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Associate Professor
School of Computer Science
University of Birmingham



**UNIVERSITY OF
BIRMINGHAM**



Lectures Times/Venues

▶ Lecturers:

- ▶ Ata Kabán: Weeks 1-5 + Consolidation Week 6
(Email: a.kaban@bham.ac.uk, Office hour: Thursdays 14:30 - 16:30)

- ▶ Hyung Jin Chang: Weeks 7-8

Email: h.j.chang@bham.ac.uk

- ▶ Miqing Li: Weeks 9-11

Office hour: Wednesday 12-1 pm / Murray
Learning Centre Office 137

▶ Lecture structure:

- ▶ Two lectures per week: Friday morning 11:00-13:00
(University Centre - UNI-Avon Room)
- ▶ Help-desk: TA Office hours
- ▶ One Zoom tutorial class per week (starting from week 2): Thursday 12:00-13:00: <https://bham-ac-uk.zoom.us/j/8499898070>
- ▶ **Assessment** = 80% exam + 20% CA

What is AI?

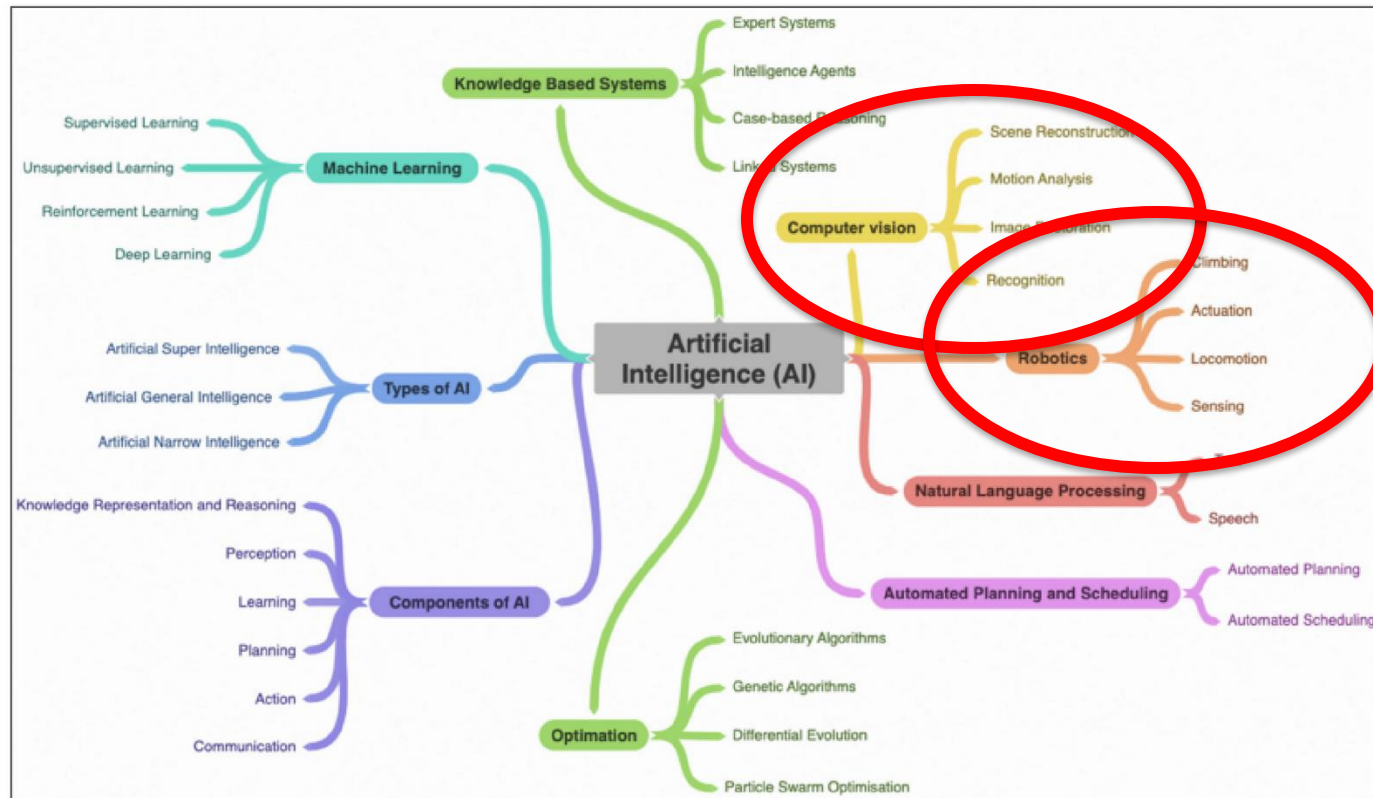


Figure: M. Regona et al., Opportunities and Adoption Challenges of AI in the Construction Industry: A PRISMA Review, JOITMC 8(1):45, 2022



- Lecturer (Hyung Jin Chang H.J.Chang@bham.ac.uk)
 - PhD @ Seoul National University, South Korea
 - Computer Vision & Robotics PostDoc @ Imperial College London before joining UoB
 - Research area: Computer Vision + Robotics
 - Teaching
 - Previous Module lead of LM Robot Vision (25024)
 - Lecturer of LM Current Topics in Artificial Intelligence and Machine Learning (32256, 32257, 34149, 37861)
 - Lecturer of LI Artificial Intelligence 2 (34255, 34256)
- Office hours
 - Room 137 of Murray Learning Centre building
 - Wednesday 12:00-13:00 pm



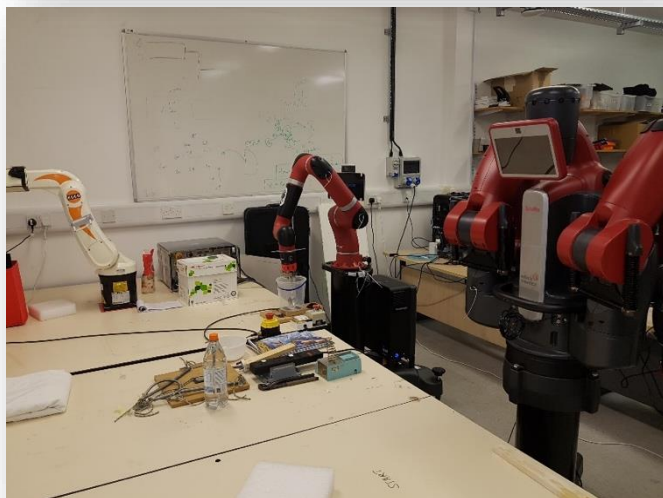
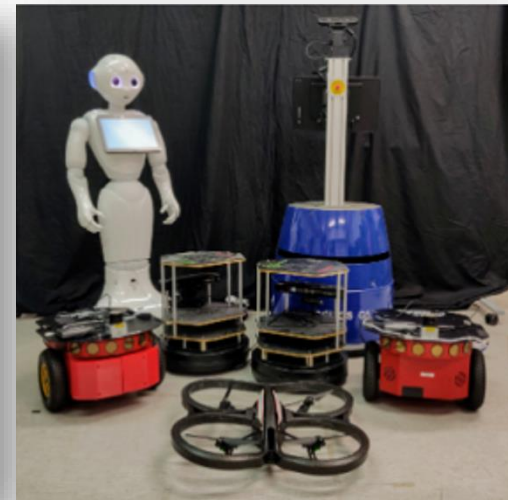
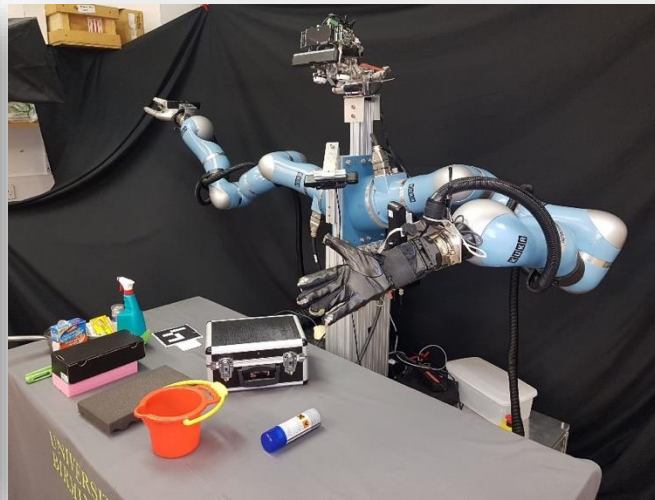
Content of this Module

- ▶ Probabilistic Artificial Intelligence (Ata):
 - ▶ Uncertainty and Estimation
 - ▶ Likelihood-based learning
 - ▶ Information theoretic quantities in learning
 - ▶ Confidence certificates: Test set bounds; Intro to train set bounds
 - ▶ Probabilistic Graphical Models
- ▶ Intro to Computer Vision & Robotics (Hyung Jin)
- ▶ Automated Planning and Scheduling (Miqing)
 - ▶ Advanced search algorithms and Constraint Satisfaction Problems



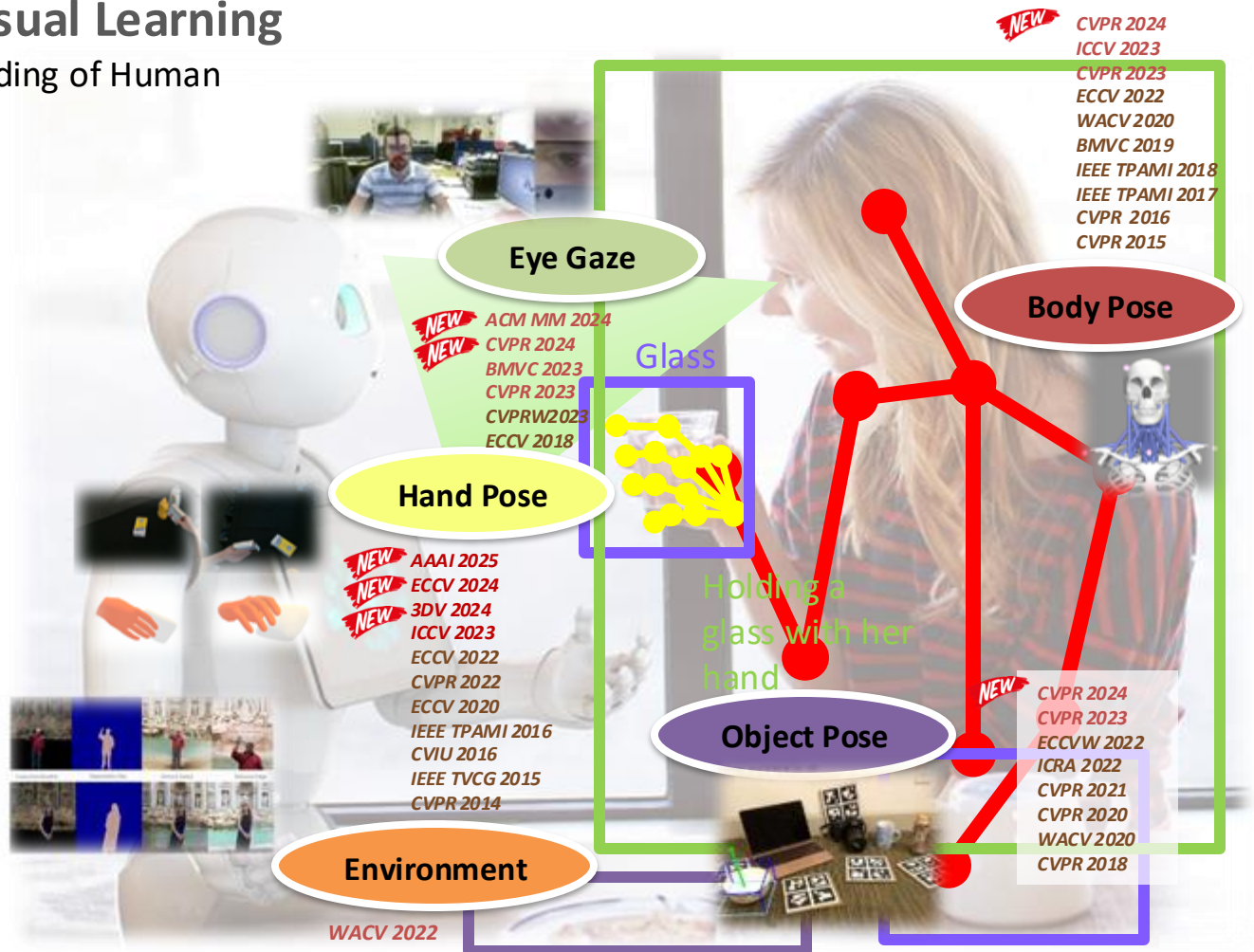


Intelligent Robotics Lab



Human-Centred Visual Learning

Comprehensive Understanding of Human
through Vision



UNIVERSITY OF
BIRMINGHAM



- 30213, 30241 Computational Vision & Imaging
- 25024 Robot Vision
- 32167 Neural Computation
- 30227 Intelligent Robotics
- 32256/32257 Current Topics in DS+AI/ML
- 35447/34147 Programming for Data Science/AI Programming

Topics covered in CV&I / Robot Vision



- Introduction to computer vision & human vision
- Image formation and colour
- Camera parameters
- Camera calibration
- Image noise
- Image filtering
- Image gradient
- Edge detection
- Matching
- Binary image analysis
- Image texture
- Image segmentation
- Fitting: voting and the Hough transform
- Image warping
- Image Stitching
- Local features: image matching and detection
- Multiple-view Geometry
- Active Imaging (3D imaging technologies)
- Face recognition
- Deep learning for computer vision I
- Deep learning for computer vision II



- **Part 1 (Week 7)**

- Introductions to computer vision
 - What is computer vision
 - Computer vision applications
- Practical tips & guidance for deep learning implementation

- **Part 2 (Week 8)**

- Challenges of computer vision tasks
- Topics of CV research
- WHAT and HOW the computer sees
- Brief overview of AI for CV (particularly neural network focused)



- **Part 1 (Week 6)**

- Introductions to computer vision
 - What is computer vision
 - Computer vision applications
- Practical tips & guidance for deep learning implementation

- **Part 2 (Week 7)**

- Challenges of computer vision tasks
- Topics of CV research
- WHAT and HOW the computer sees
- Brief overview of AI for CV (particularly neural network focused)



Google

AI



Q All

News

Images

Videos

Books

More

Tools



Forbes
Where Does AI Go Next?



Plat.AI
Build an AI ...



Simplilearn
Artificial Intelligence ...



TIME
Using AI to Its Fullest Human Potential ...



Simplilearn
7 Types of Artificial Intelligence That ...



Insurt
Insurtech



Emarsys
What Is Real Artificial Intelligence ...



Springboard
Is AI Hard To Learn? A Guide To Getting ...



Nature
Artificial intelligence



The Hindu Business Line
Leveraging AI for tackling corruption ...



Camino Partners
AI impact recruitment marketing ...



RAID Confe
AI's balancing



McKinsey
Artificial Intelligence Insights ...



gmailfoundry.ac.uk
Greater Manchester AI | Gre...



The World Economic Forum
A 5-step guide to scale responsible AI ...



The Times of Israel
All-in-one chips seen boosting computer ...

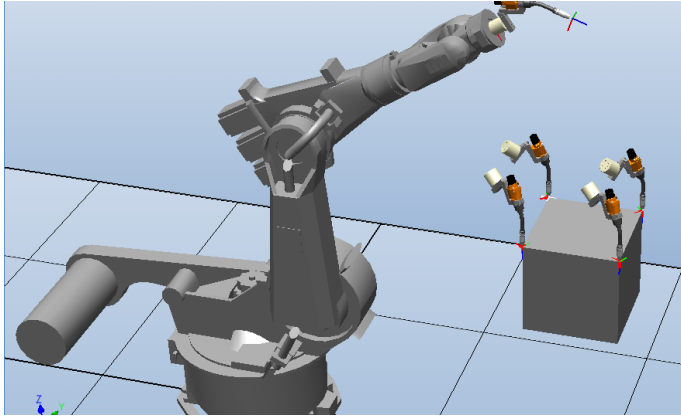


Corporate Finance Institute
Artificial Intelligence (AI) - Overview ...

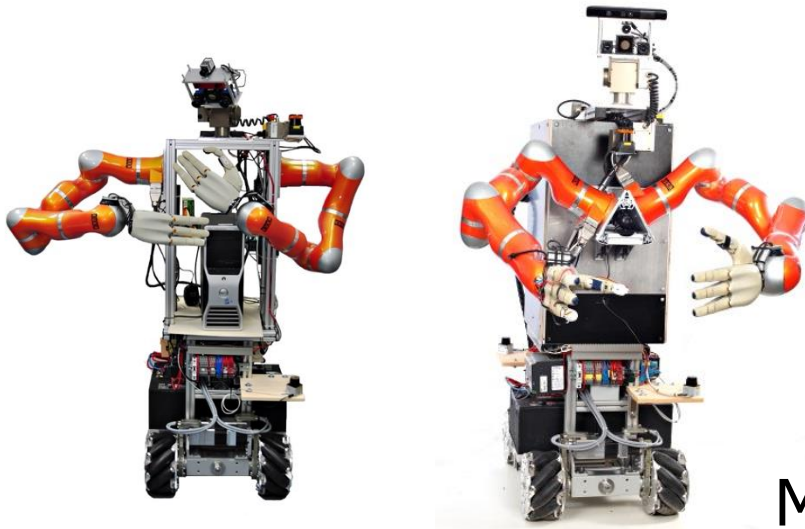


Embodied **AI** = Robot

Robots

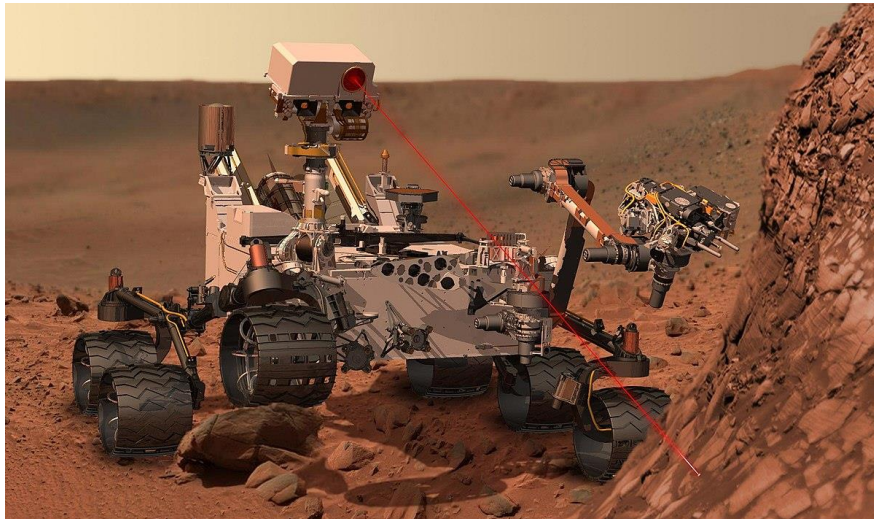
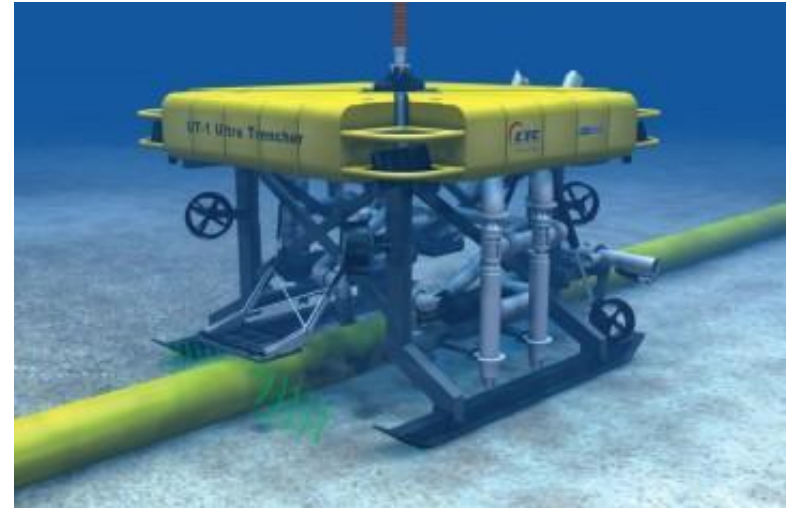
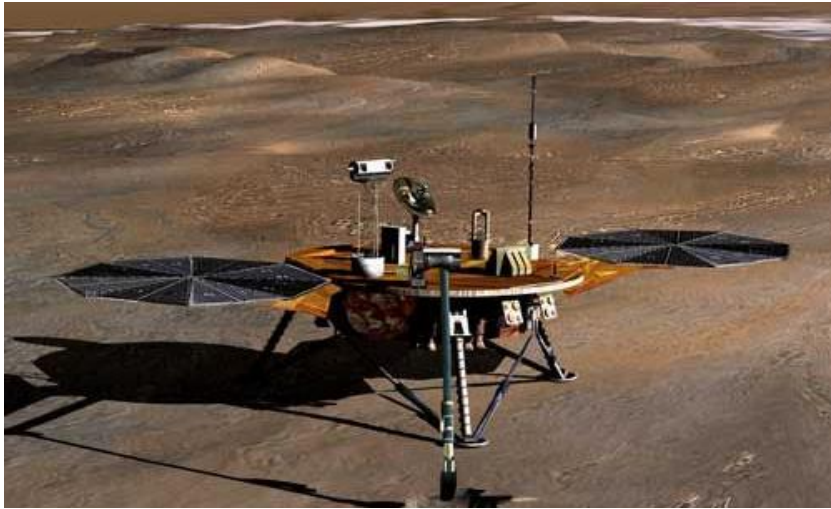


Industrial robots



Mobile robots in warehouses

Mars robot, underwater robots



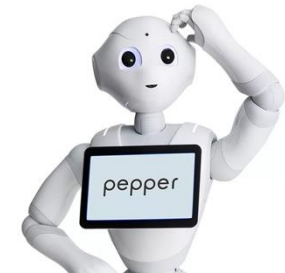
UAV / Drone



Autonomous Vehicle



Humanoid robots



A **humanoid robot** is a **robot** with its body shape built to resemble the human body. The design may be for functional purposes, such as interacting with human tools and environments, or, for example, for experimental purposes, such as the study of locomotion.

Why Humanoids?

Key Reasons for Human-like Robots

- **Human Ergonomics:** Humanoid robots are designed with human-like structure to enhance compatibility with tools, furniture, and environments intended for humans.
- **Seamless Integration in Human Environments:** They enable tasks in shared spaces using the same furniture, tools, and workspaces, reducing the need for modifications.
- **Enhanced Natural Interaction:** Humanoids can mimic human gestures and movements, fostering more intuitive interactions and minimizing psychological barriers.

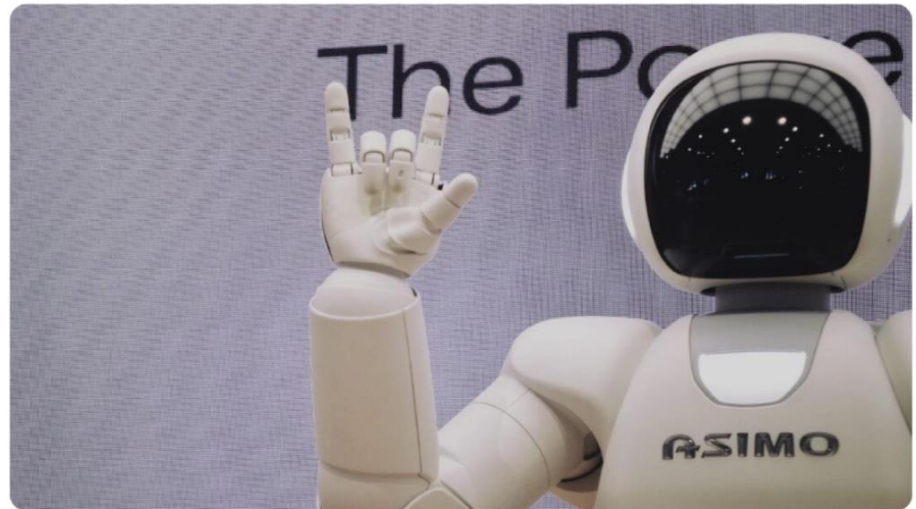
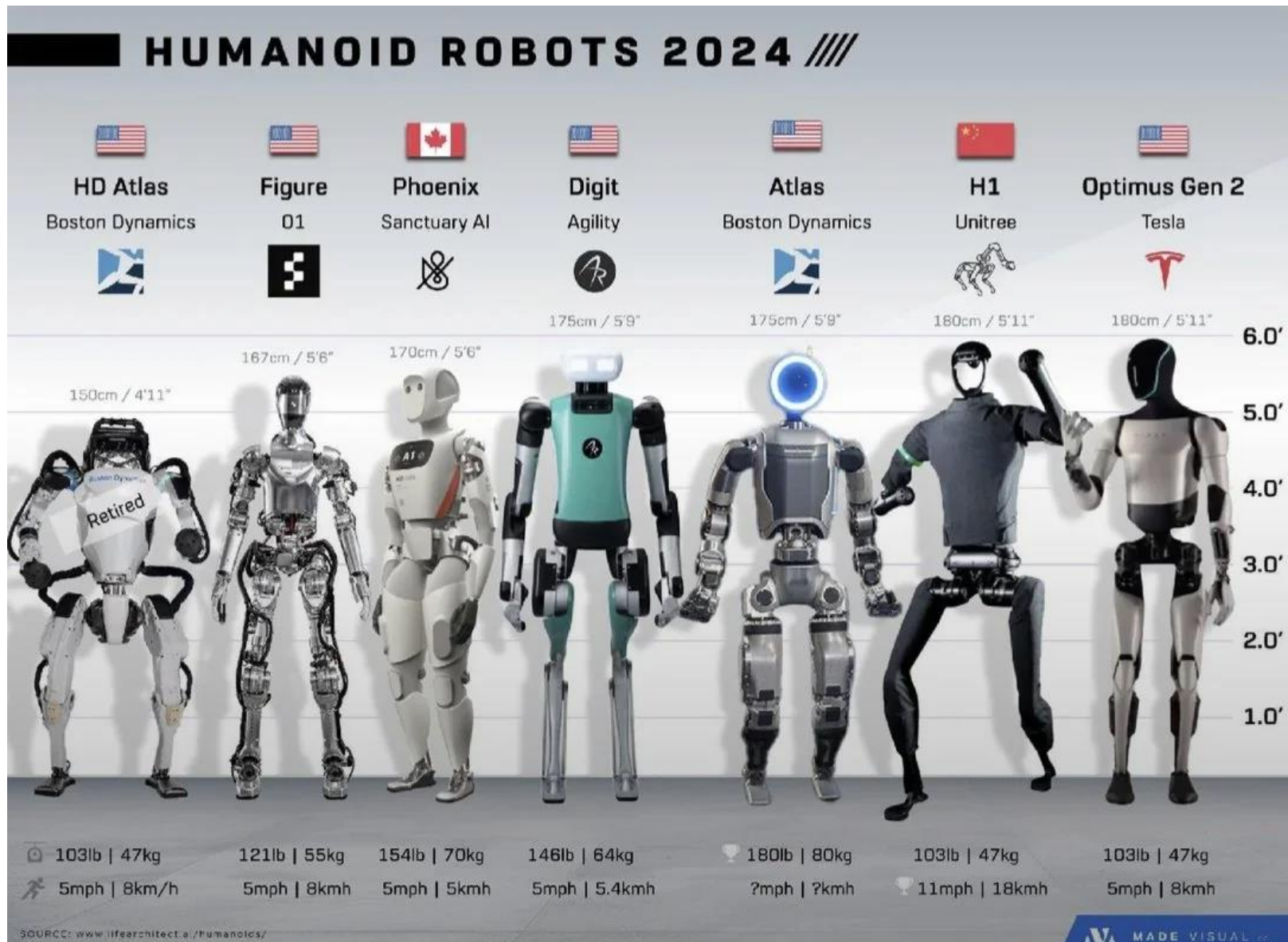
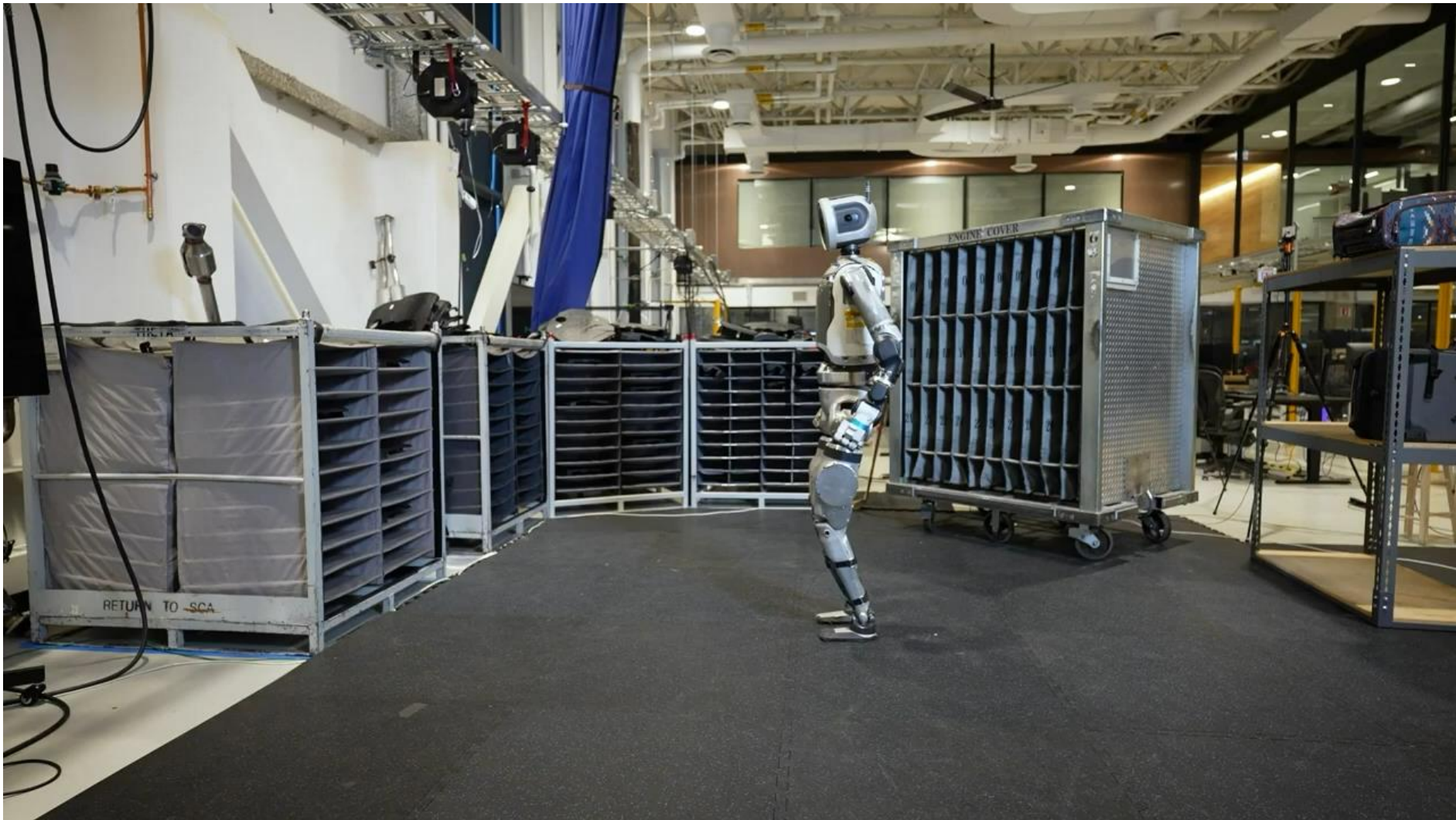


Photo by Possessed Photography on Unsplash





<https://www.youtube.com/watch?v=oL5YNtDUQXU>



https://www.youtube.com/watch?v=F_7IPm7f1vI



<https://www.youtube.com/watch?v=DrNcXgoFv20>



<https://www.youtube.com/watch?v=yEOjGKA6Kvo>



Learning Mechanisms for Robots

Training Techniques for Advanced Robots



Direct and Remote-Controlled Learning

Robots can learn by receiving direct commands or via remote control, allowing operators to guide them through tasks and refine their responses.



Simulation-Based Training

Using virtual simulations, robots practice tasks in controlled digital environments, accelerating learning cycles and reducing risks.



Human-Centered Visual Learning

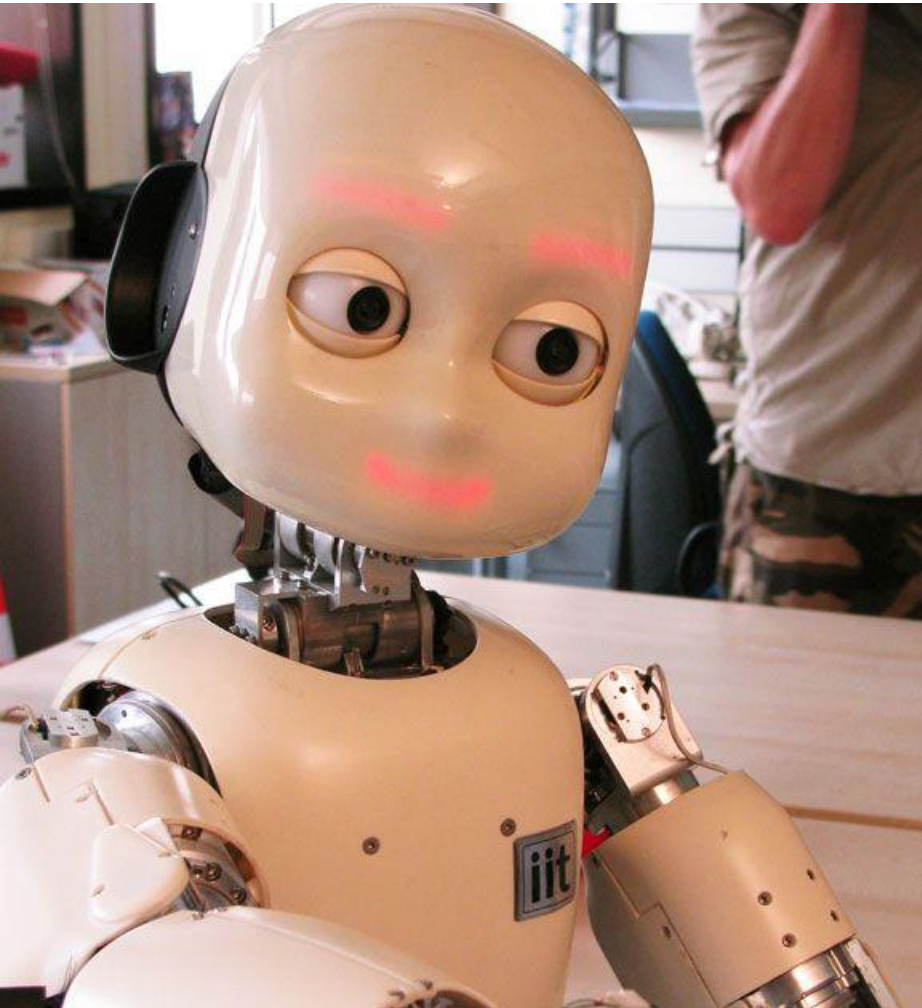
By observing human movement, hand-object dynamics, and gaze, robots learn intuitive actions and more complex interactions.

Robot Vision

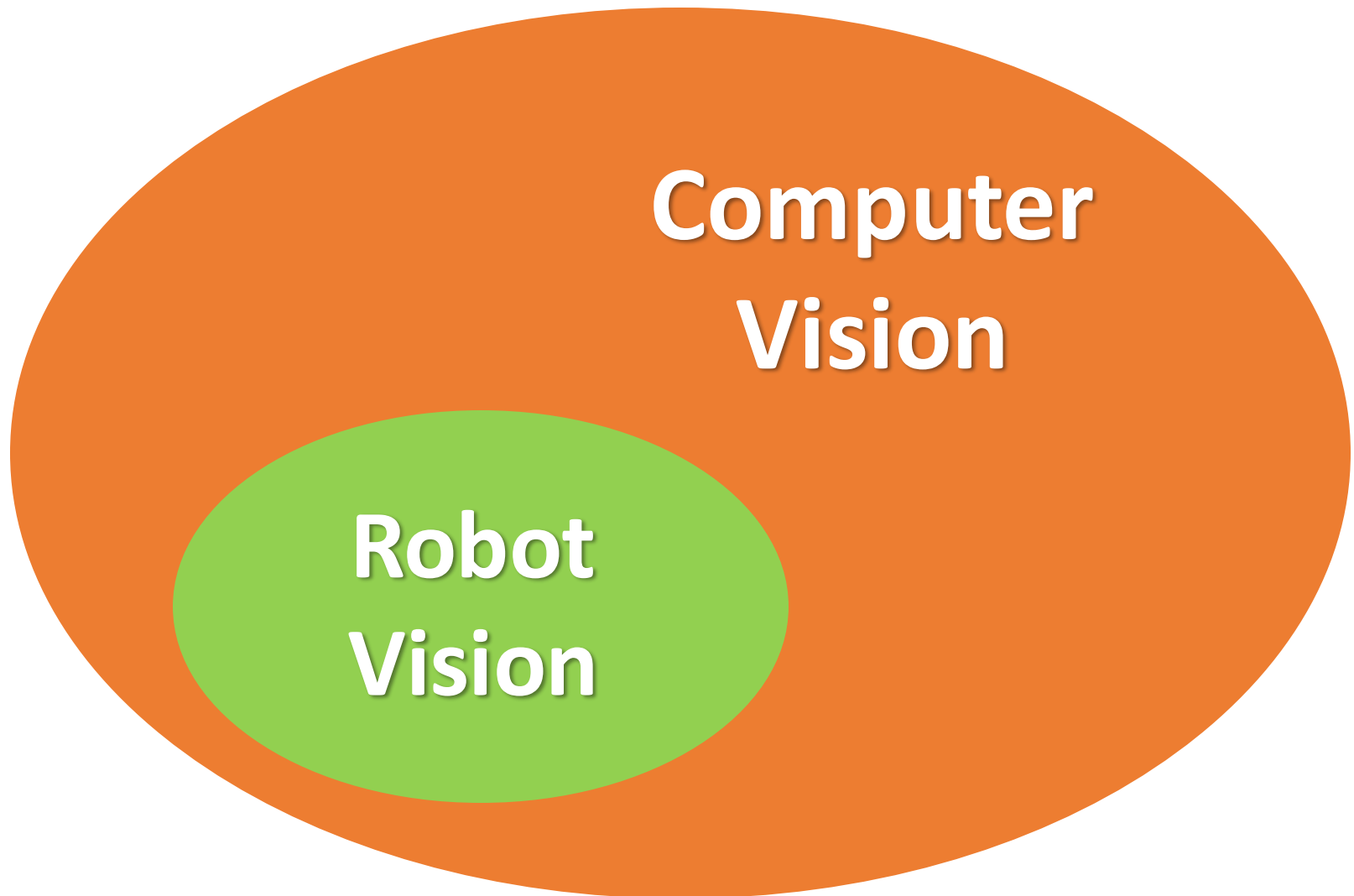


Terminator 1984

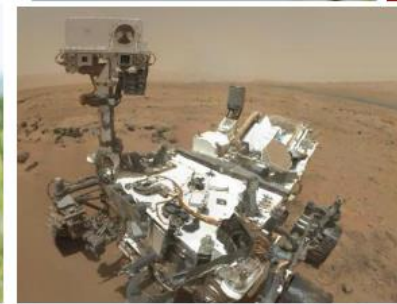
Robot Vision?



Robot Vision? Computer Vision!



Welcome to Computer Vision!



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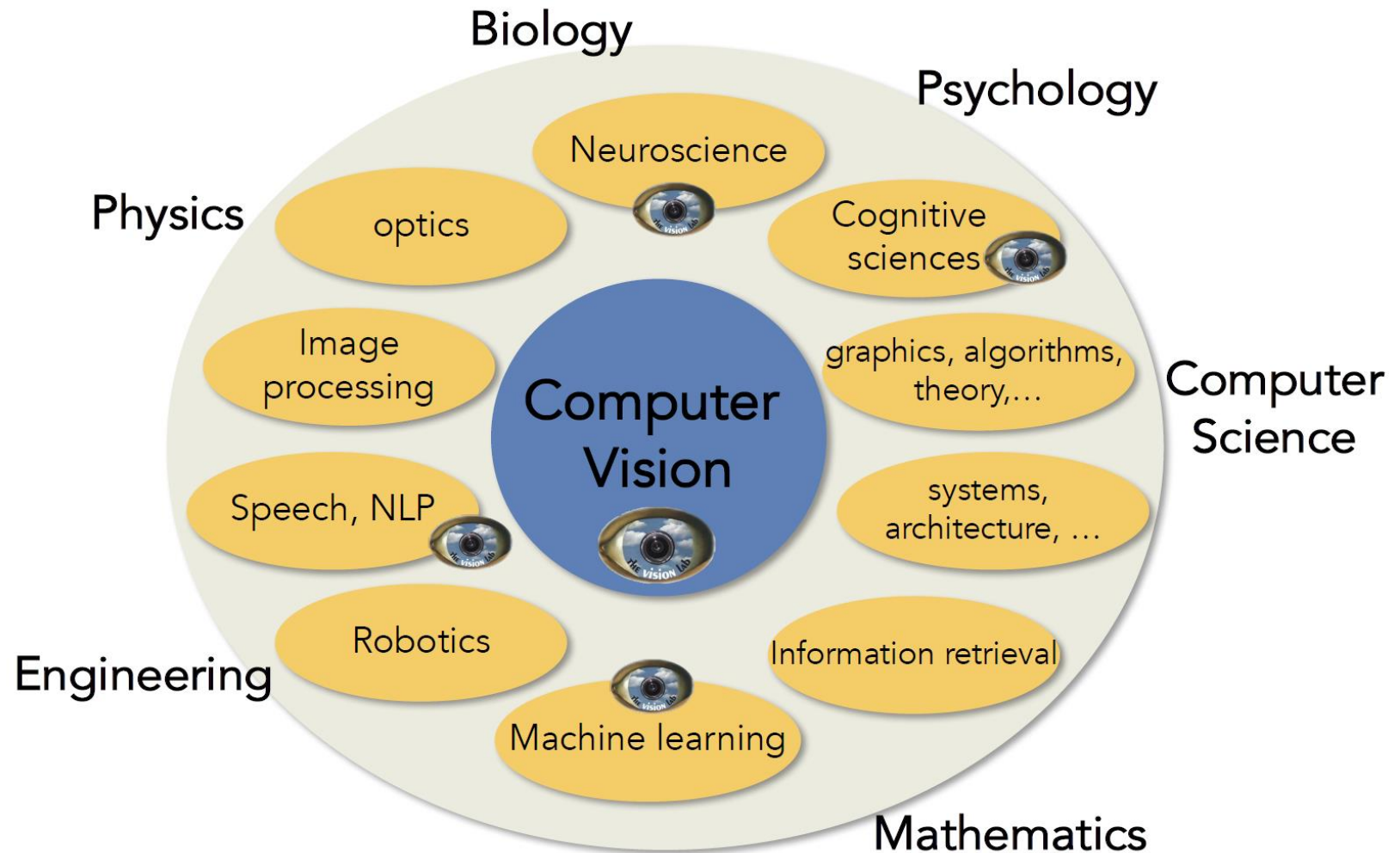
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Slide credit: Fei-Fei Li



What is Computer Vision?

Multidisciplinary field



Slide credit: Fei-Fei Li



- Automatic understanding of images and video
 1. Computing properties of the 3D world from visual data
(*measurement*)

1. Vision for measurement



Real-time stereo



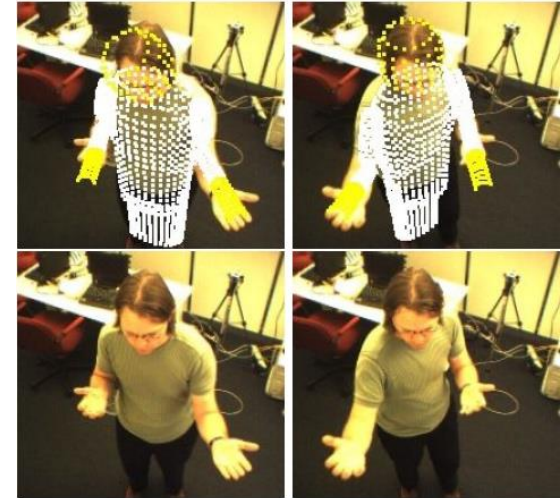
Wang et al.

Structure from motion



Snaveley et al.

Tracking



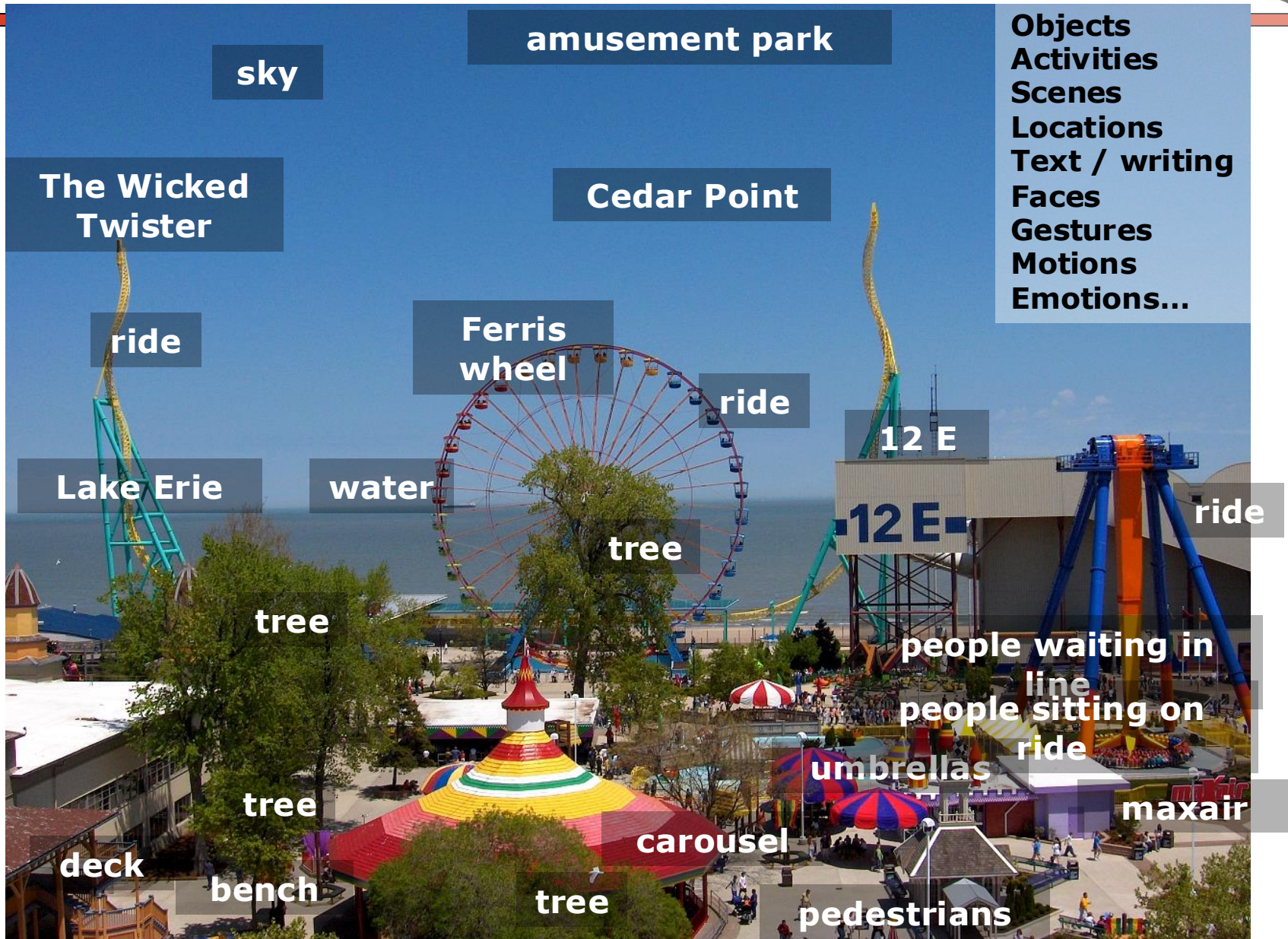
Demirdjian et al.

Slide credit: Kristen Grauman



- Automatic understanding of images and video
 1. Computing properties of the 3D world from visual data
(*measurement*)
 2. Algorithms and representations to allow a machine to recognise objects, people, scenes, and activities
(*perception and interpretation*)

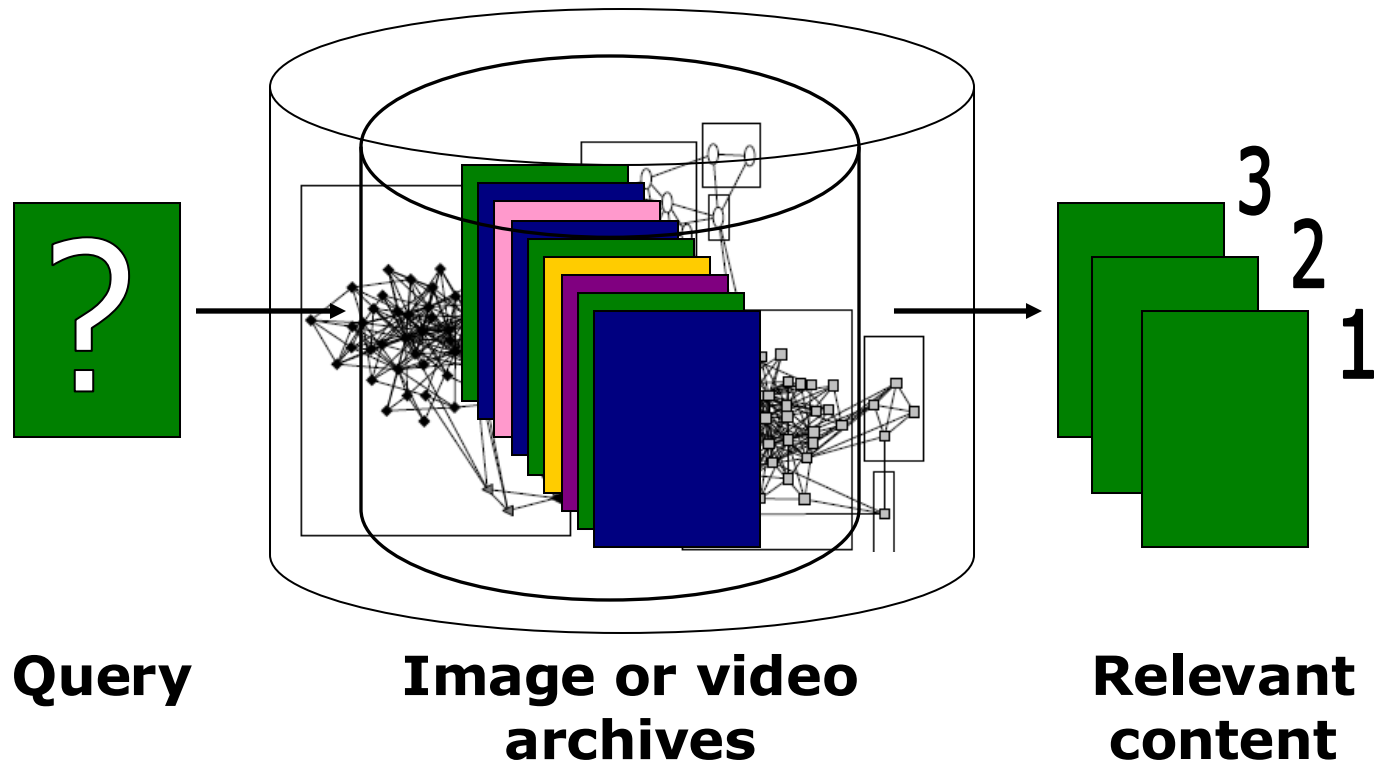
2. Vision for perception, interpretation





- Automatic understanding of images and video
 1. Computing properties of the 3D world from visual data
(*measurement*)
 2. Algorithms and representations to allow a machine to recognise objects, people, scenes, and activities
(*perception and interpretation*)
 3. Algorithms to mine, search, and interact with visual data
(*search and organisation*)

3. Visual search, organisation



Slide credit: Kristen Grauman



- Automatic understanding of images and video
 1. Computing properties of the 3D world from visual data
(*measurement*)
 2. Algorithms and representations to allow a machine to recognise objects, people, scenes, and activities
(*perception and interpretation*)
 3. Algorithms to mine, search, and interact with visual data
(*search and organisation*)
 4. Reconstructing 2D/3D visual environments and objects
(*visualisation*)



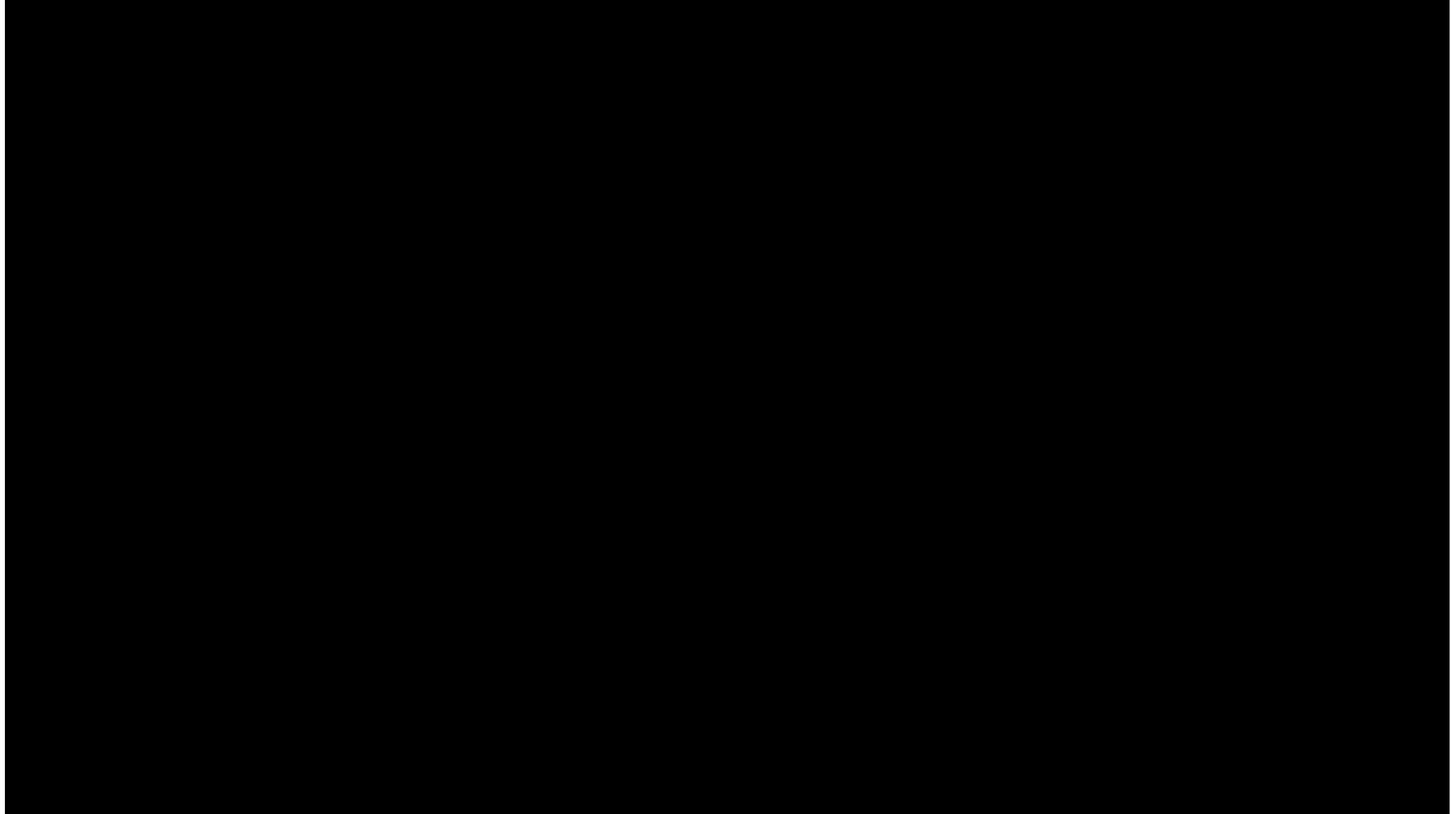
Computer Vision Applications

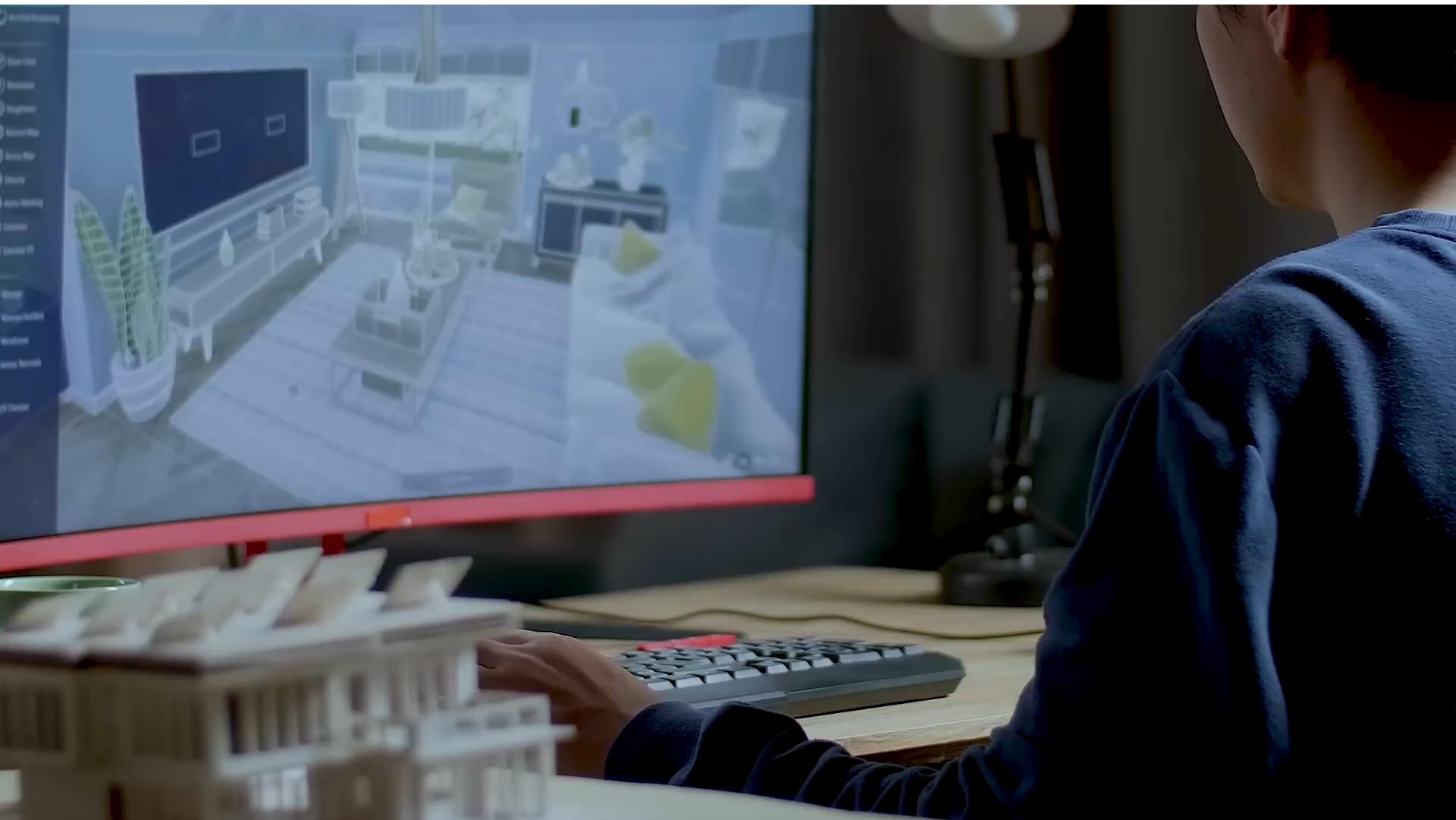
Industrial Robot + Vision



Robot @ Ocado





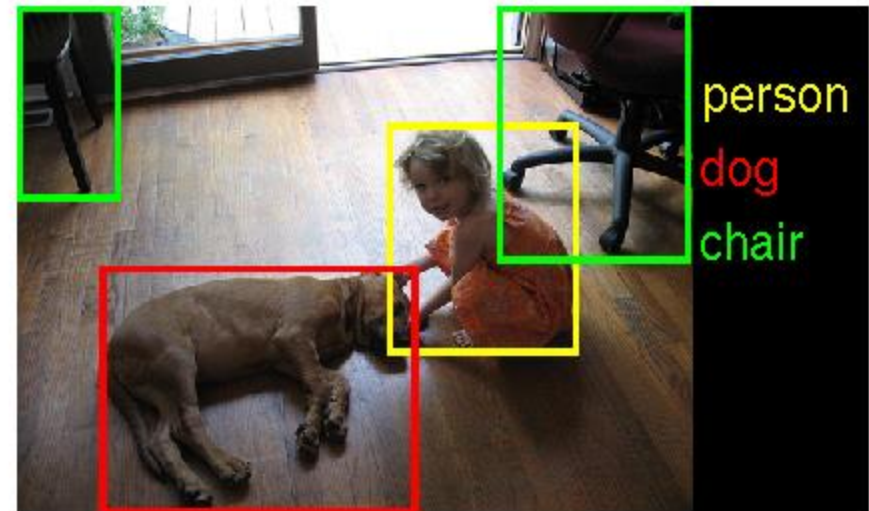


<https://www.youtube.com/watch?v=cZgsR7KdvBQ>

Image Classification and Object Detection



Image classification



Object detection

Object Detection & Recognition

The background of the slide features a dark teal gradient with a network of glowing blue nodes connected by thin lines, resembling a neural network or a data structure. The text "YOLO v2" is prominently displayed in the center in a large, white, sans-serif font.

YOLO v2

<http://pureddie.com/yolo>

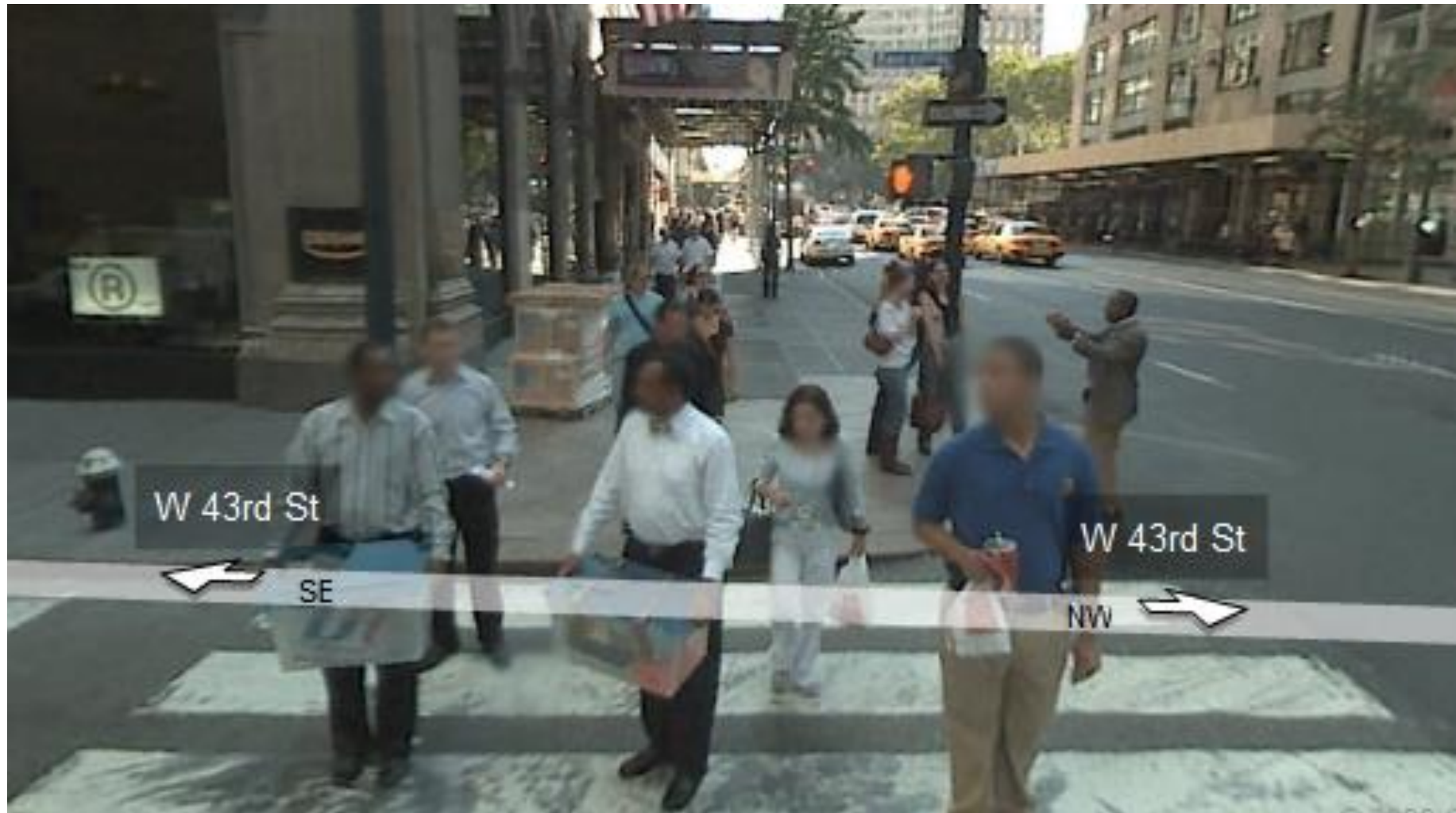
Tracking & Recognition



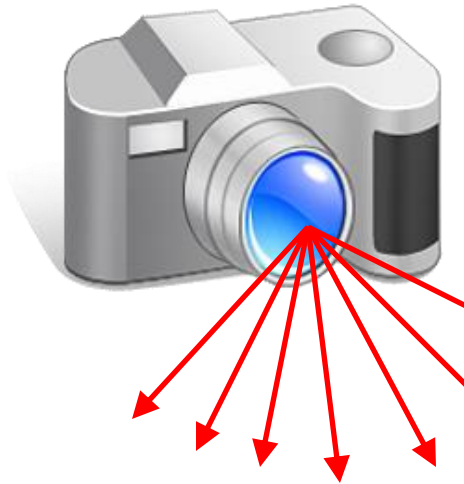
Face Detection



Face Detection for Privacy Protection



Mosaics



...



image from S. Seitz

Obtain a wider angle view by combining multiple images.

Let's Try!

- **How to take a panorama with your iPhone or iPad**
 - Launch the **Camera** app on your **iPhone** or iPad.
 - Swipe left twice to change modes to Pano.
 - Tap the arrow button to change the capture direction, if desired.
 - ...
 - Tap the shutter button to start taking a **panoramic** photo.
- **Android**
 - Select **Panoramic** Mode. Launch the camera on your **Android phone** or tablet. Then tap the "MODE" button. ...
 - **Take Panoramic** Shot with **Android** Camera. Tap the shutter button.
 - Save and Store **Panoramic** Shot to **Android Device**. Tap the shutter again to end photographing.

Image Colourisation

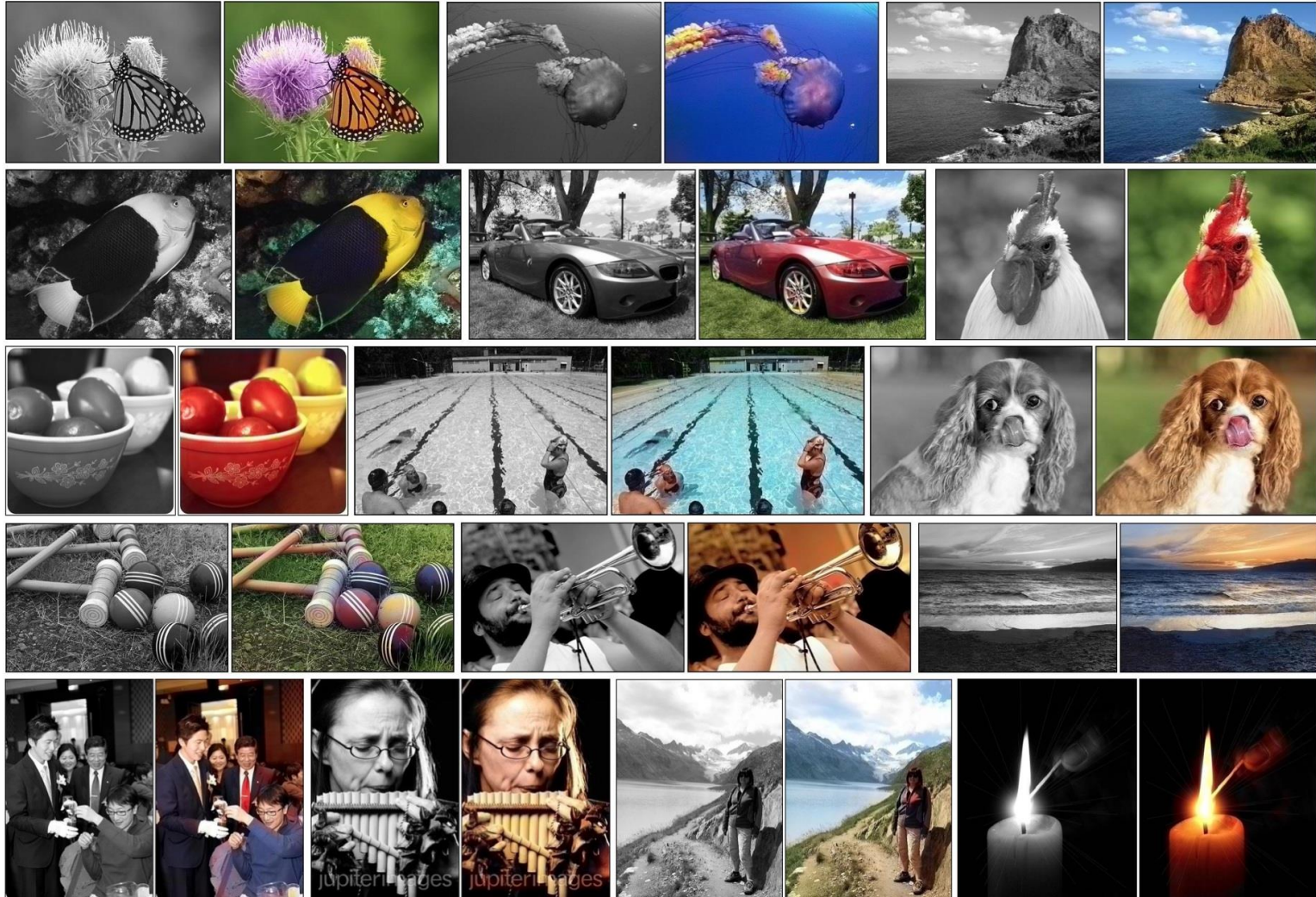


Image Color

Google Play Games Apps Movies & TV Books Kids



Store Mac iPad iPhone Watch AirPods TV & Home Entertainment Accessories Support

App Store Preview

Open the Mac App Store to buy and download apps.



Colorize - Color to Old Photos

Black and White Colorization

Photomyne LTD

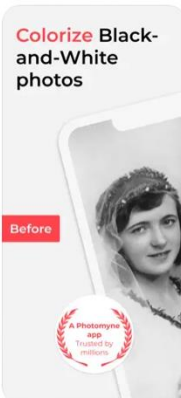
Designed for iPhone

★★★★★ 4.7 • 9.5K Ratings

Free - Offers In-App Purchases

[View in Mac App Store](#)

iPhone Screenshots



Brought to you by the developer that built the smartest photo scanner app, Photomyne is proud to now also offer an AI-powered B&W colorization app that will blow you away. It's one of the most accurate old photo colorization apps - the results will amaze you.

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Colorize Images

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Contains ads - In-app purchases



3.9★

13.6K reviews

1M+

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Everyone 0

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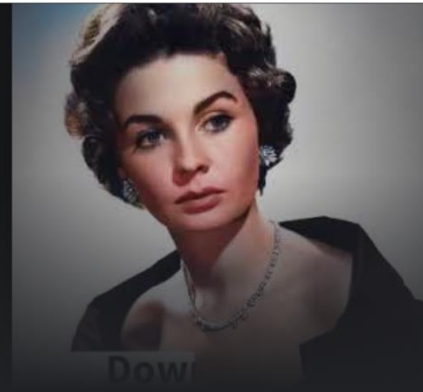


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▶ Trailer



About this app

Bring new life to your old black and white photos with Colorize Images. Our AI-powered app uses advanced machine learning algorithms to automatically add vibrant colors to your grayscale images, giving them a fresh and modern look.

With a simple and intuitive interface, Colorize Images makes it easy to transform your photos into colorful works of art. Select a photo from your device's gallery or share it from another app, and let our app do the rest. Save your colorized photos to your gallery, or share them with friends and family on social media.

Developer contact

Similar apps

- Colorize - Color to Old Photos
Photomyne Ltd.
4.4 ★
- AI Enlarger for Photo & Anime
Vertexshare Software Ltd
4.4 ★
- Print adjuster
kremi151
4.7 ★
- Sketch on PDF
Applicationsprogramvara
3.9 ★



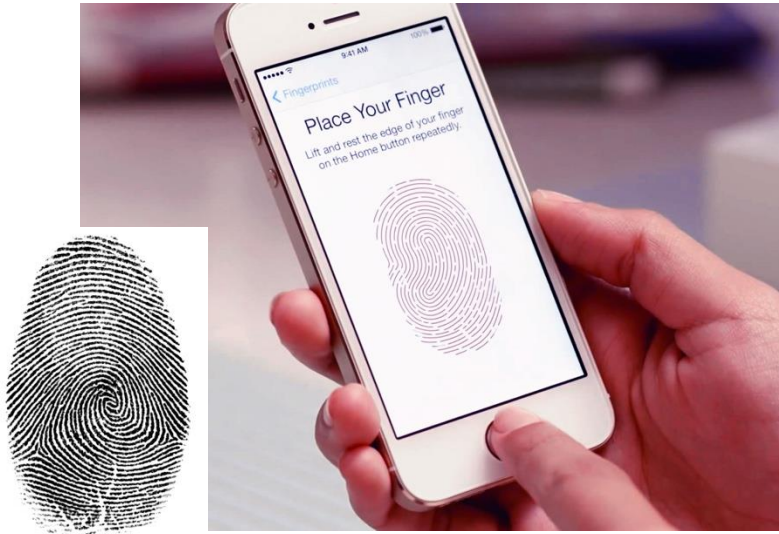
June 1937



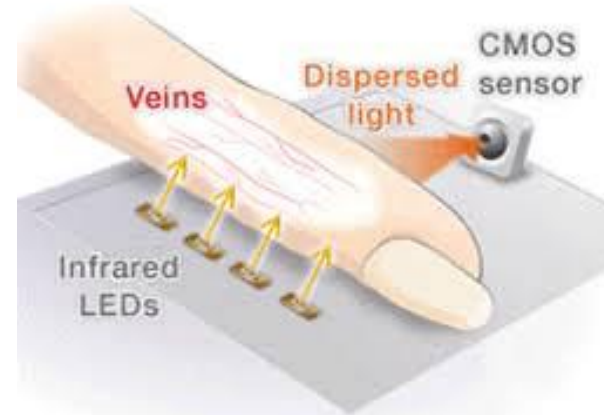
Berry Field, June 1909



Hamilton, 1936



Fingerprint scanners



Finger vein recognition



Face recognition systems



Palm vein recognition

Apple's Face ID

A grayscale image of a man's face, centered in the frame. The face is slightly tilted and has a neutral expression. The background is dark and textured.

**HOW FACE ID
ACTUALLY WORKS**

<https://www.youtube.com/watch?v=g4m6StzUcOw>

Interaction in Augmented Reality





<https://www.youtube.com/watch?v=IY4x85zqoJM>

Apple Vision Pro



Apple Vision Pro Avatar FaceTime

 **sid** 
@immasiddtweets

Imagine spending \$3499 on an Apple Vision Pro only for it to make you look like a video game character from a decade ago in FaceTime calls 😂



 sid



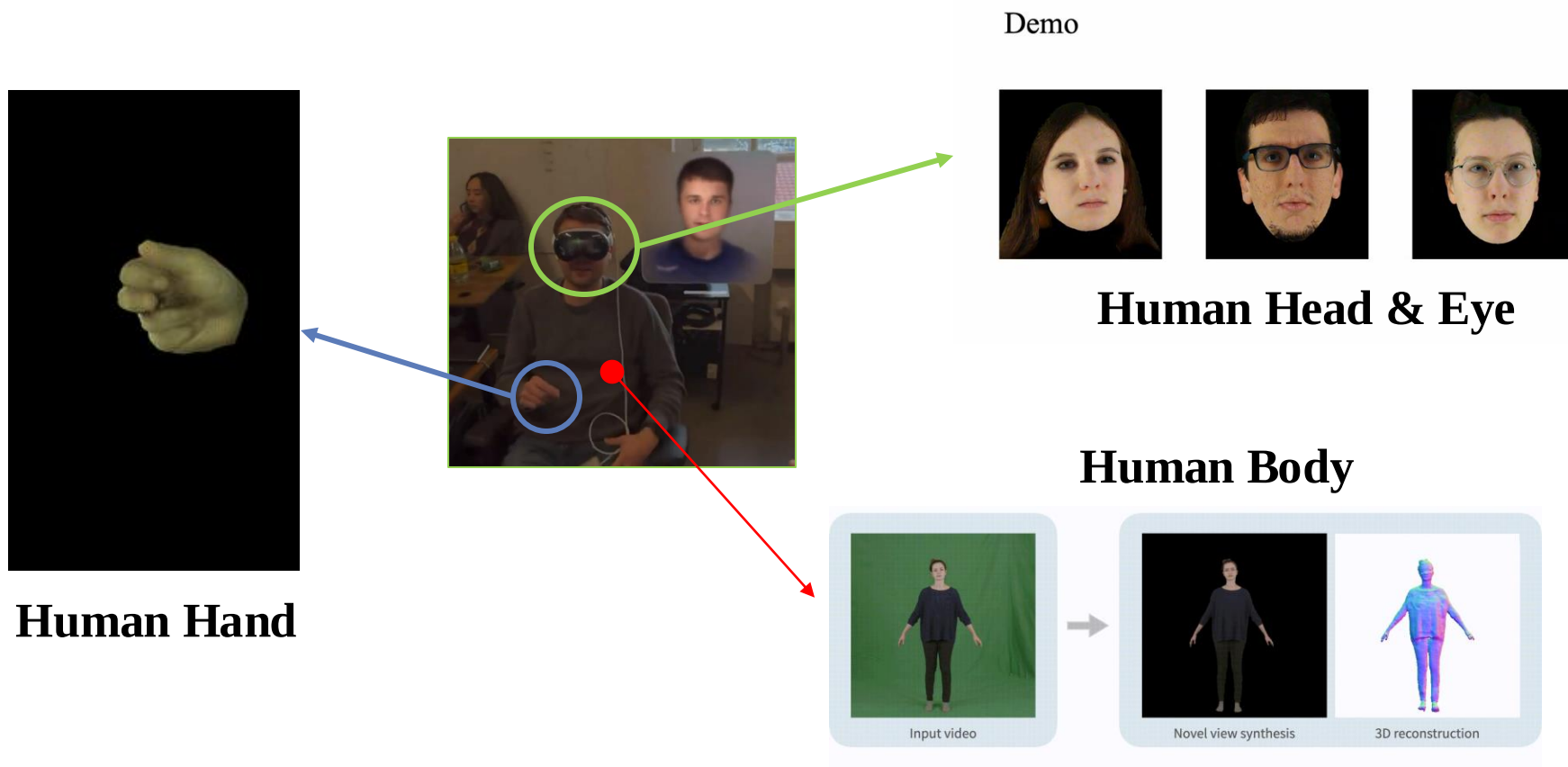
 BHD



 ng



Our Group on **Virtual Human Creation**



Autonomous Vehicle



3D Reconstruction from Photo Collections



Colosseum, Rome, Italy



San Marco Square, Venice, Italy



Q. Shan, R. Adams, B. Curless, Y. Furukawa, and S. Seitz, [The Visual Turing Test for Scene Reconstruction](#), 3DV 2013

Slide credit: Svetlana Lazebnik

- <https://www.google.com/maps/@52.4461709,-1.9313211,269a,35y,7.73h,56.95t/data=!3m1!1e3>



Neural Rendering

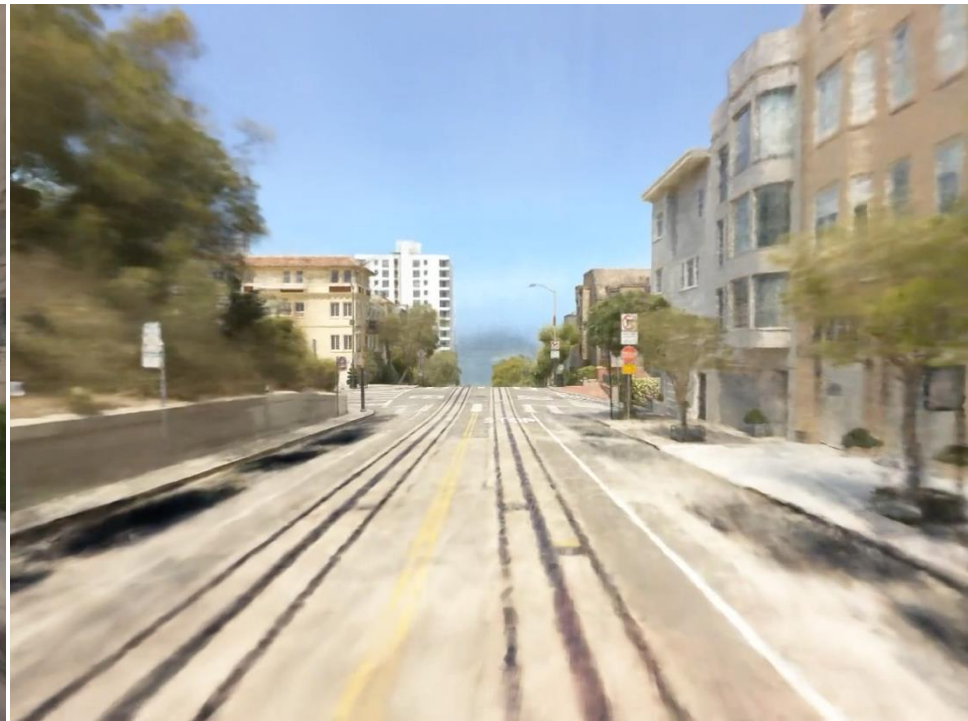


<https://www.youtube.com/watch?v=DJ2hcC1orc4>

Neural Rendering



- City scale rendering - Block NERF



<https://waymo.com/research/block-nerf/>

<https://datagen.tech/guides/synthetic-data/neural-radiance-field-nerf/>

PhysGaussian: Physics-Integrated 3D Gaussians for Generative Dynamics

Tianyi Xie^{1*}, Zeshun Zong^{1*}, Yuxing Qiu^{1*}, Xuan Li^{1*},
Yutao Feng^{2 3}, Yin Yang³, Chenfanfu Jiang¹

UCLA



浙江大学
ZHEJIANG UNIVERSITY

THE
UNIVERSITY
OF UTAH

* Equal contributions

[with audio 🎧]

<https://www.youtube.com/watch?v=V96GfcMUH2Q>

From 2D to 3D



S. Saito et al., “PIFuHD: Multi-Level Pixel-Aligned Implicit Function for High-Resolution 3D Human Digitization”, CVPR 2020

From 2D to 3D



S. Saito et al.. “PIFuHD: Multi-Level Pixel-Aligned Implicit Function for High-Resolution 3D Human Digitization”, CVPR 2020

Reconstruct 3D Humans and Environments



Input video



HuMoR with our context



Novel view



Pavlakos et al. ECCV 2022



One Shot 3D Photography

Johannes Kopf
Ocean Quigley
Josh Patterson
Matthew Yu
Peter Vajda

Kevin Matzen
Francis Ge
Jan-Michael Frahm
Peizhao Zhang
Ayush Saraf

Suhib Alsisan
Yangming Chong
Shu Wu
Zijian He
Michael Cohen



Photo Wake-Up: 3D Character Animation from a Single Photo

Chung-Yi Weng¹, Brian Curless¹, Ira Kemelmacher-Shlizerman^{1,2}

¹University of Washington

²Facebook, Inc.

Video Generation

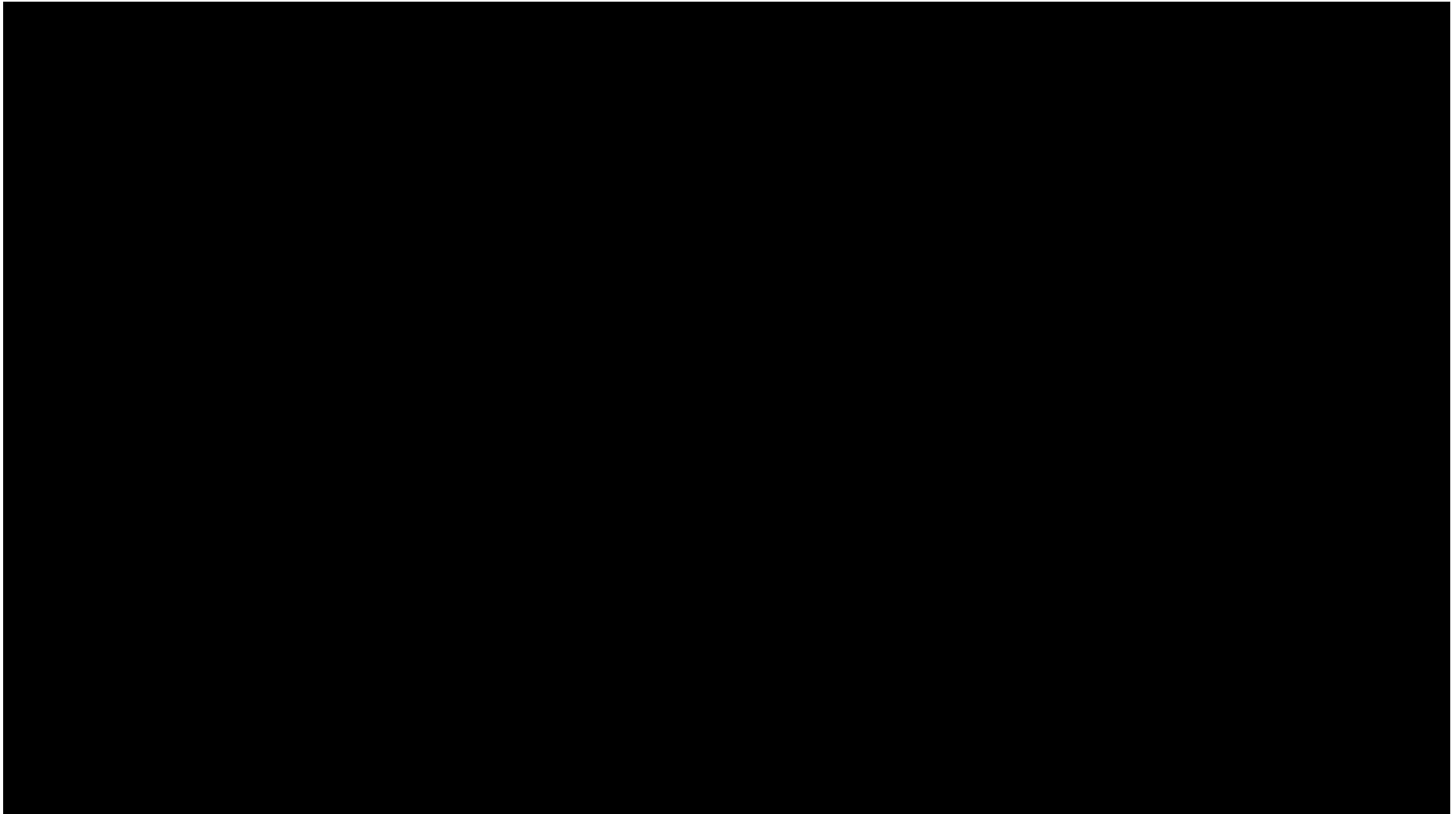


<https://www.youtube.com/watch?v=anmuklFtu8U>



openai.com/sora

Video Generation for Physical AI



<https://www.youtube.com/watch?v=9Uch931cDx8>



- As image sources multiply, so do applications
 - Relieve humans of boring, easy tasks
 - Enhance human abilities
 - Advance human-computer interaction, visualisation
 - Perception for robotics / autonomous agents
 - Organise and give access to visual content

Innovative Applications of Computer Vision



- Verifying financial documentation
- Assessing the health of ocean ecosystem
- Streamlining supply chain operations
- Optimizing Retail Inventory Management And Marketing
- Automating QA For Manufacturers
- Automating Retail Checkout
- Creating Knowledge Bases
- Converting Paper Documents Into Digital Files
- Spotting Phishing Content
- Predicting Potential Security Threats
- Assessing The Appeal Of Real Estate
- Monitoring The Health Of Crops
- Enabling Visual Search
- Conserving Wildlife And Natural Resources
- Making Workplaces Safer
- Monitoring Green Energy Generation
- Hyper-Personalizing Shopping Selections
- Analyzing Medical Images
- Screening Job Candidates
- Powering Immersive AR

<https://www.forbes.com/sites/forbestechcouncil/2024/02/16/take-a-look-at-innovative-applications-of-computer-vision/amp/>



- The core of AI lies in computer vision, which has recently driven its drastic success.
- It is a fascinating and challenging field, with plenty of useful and surprising applications.
- Computer vision has opened up new opportunities, with the emergence of new companies, startups, and more.

Summary



- Computer Vision is super HOT!

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Any question?

